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Media Literacy Interventions Improve Resilience to Misinformation: A Meta-Analytic Investigation of Overall Effect and Moderating Factors

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Abstract

The widespread dissemination of misinformation has become a global concern. A recommended solution is to improve people's ability to discern true from false information through appropriate media literacy education programs. This meta-analysis quantitatively synthesized the results of 49 experimental studies ($N = 81,155$) that examined the efficacy of media literacy interventions in mitigating misinformation. This study finds that media literacy interventions generally improve resilience to misinformation ($d = 0.60$). Specifically, the interventions reduce belief in misinformation ($d = 0.27$), improve misinformation discernment ($d = 0.76$), and decrease misinformation sharing ($d = 1.04$). Moreover, media literacy interventions have stronger effects (1) when multiple sessions rather than a single session are implemented, (2) in high (versus low) uncertainty avoidance cultures, and (3) among college students than among adults recruited from online crowdsourcing platforms (e.g., Amazon Mechanical Turk). These findings enrich our understanding of inoculation theory and provide valuable guidance for the design of future media literacy intervention programs.

Keywords: misinformation, media literacy, intervention program, cultural dimension

Media Literacy Interventions Improve Resilience to Misinformation: A Meta-Analytic Investigation of Overall Effect and Moderating Factors

“At a time of rampant misinformation, climate denial, and attacks on human rights, we need education systems that distinguish fact from conspiracy, instill respect for science, and celebrate humanity in all its diversity.”

António Guterres, the United Nations Secretary-General’s opening remarks at the
Transforming Education Summit

As quoted above, rampant misinformation, disinformation, and fake news in digital media environments have become a global concern and affect many areas critical to human welfare. Examples of detrimental consequences include increased vaccine resistance (Lee et al., 2022) and delayed climate action (van der Linden et al., 2017). In this essay, “misinformation”—which refers to “inaccurate or misleading information” (Yu et al., 2023, p. 2242)—is used as a general term that encompasses disinformation (i.e., intentionally misleading information), fake news (i.e., incorrect information in news format), and rumors (i.e., unauthenticated information). The ongoing fight against misinformation requires a joint effort from governments, news organizations, social media platforms, and the public. Possibly the most sustainable remedy for the adverse consequences of misinformation dissemination is to equip the public with the ability to “distinguish fact from conspiracy,” or specifically, to enhance media literacy.

A growing number of studies have used experimental methods to examine the effectiveness of media literacy interventions in combating misinformation, and mixed findings have been reported regarding various misinformation-related outcomes. For example, empirical studies have shown that media literacy interventions decreased belief in misinformation and intention to share misinformation (Adjin-Tettey, 2022; Apuke et al., 2022; Hwang et al., 2021; Scheibenzuber et al., 2021). However, some studies found that

exposure to tweets involving a news literacy intervention had little impact on reducing misperceptions about food safety (Vraga et al., 2021; 2022), whereas other studies showed that digital media literacy interventions increased older adults' ability to discern fake news from real news (Moore & Hancock, 2022). Given these inconclusive results, it is essential to aggregate empirical findings across studies and unravel the boundary conditions under which media literacy interventions play a positive, negative, or insignificant role in countering misinformation.

Jeong and colleagues (2012) conducted a meta-analysis of the general outcomes of media literacy interventions in 2012, and they found that media literacy interventions had a significant positive effect ($d = 0.37$) across domains. In the past decade, additional meta-analyses have been conducted on the role of media literacy interventions in specific domains such as unhealthy behaviors (e.g., alcohol use, smoking, and eating disorders; Xie et al., 2019) and risky sexual behaviors (Vahedi et al., 2018). To date, no meta-analysis on the relationship between media literacy interventions and misinformation-related outcomes has been undertaken. To address this research gap, we conducted this meta-analysis to reveal the direction and magnitude of the overall effect and uncover the moderating factors that amplify, lessen, or nullify the effect. The findings illuminate the nuances in the role of media literacy interventions in the fight against misinformation and provide valuable insights for enhancing the effectiveness of media literacy education for policymakers, educators, and news agencies.

Media Literacy: A Complex Concept

A common definition of media literacy, formalized at the 1992 National Leadership Conference on Media Literacy, is “the ability to access, analyze, evaluate, and communicate messages in various forms” (Aufderheide, 1993, p. 6). Based on this definition, scholars have further defined media literacy and its pivotal role as “a critical-thinking skill that enables

audiences to decipher the information that they receive through the channels of mass communications” (Silverblatt et al., 1997, p. 48). Therefore, media literacy as an ability or skill is naturally linked to information discernment and the corresponding communicative actions. Media literacy can thus empower people “to be conscious about what’s going on around you and not be passive and, therefore, vulnerable” (Thoman, 1999, p. 50).

Potter (2022) conceptualized media literacy as a multi-faceted construct comprised of “skills, knowledge structures, and personal focus” (p. 15). According to Potter (2022), the level of media literacy is determined by the complex interplay of three dimensions: the information and skills (dimension 1) that people learn are organized into knowledge structures (dimension 2) in their minds, and personal focus (dimension 3) provides the mental motivation that directs the critical processing of media messages. Echoing the concept of the empowering role of media literacy (Hobbs, 2013; Thoman, 1999), Potter (2022) asserted that “increasing your level of media literacy gives you more power to control media exposures and their eventual effect on you” (p. 30). Media literacy interventions have been found to be effective in mitigating the harmful consequences of media exposure that include violence (Bickham & Slaby, 2012), unhealthy behaviors (e.g., alcohol use, smoking, and eating disorders; Xie et al., 2019), and risky sexual behaviors (Vahedi et al., 2018).

The concept of media literacy has been evolving with the transformation of the media ecosystem. The early definitions of media literacy born in the mass media environment (e.g., Aufderheide, 1993; Silverblatt et al., 1997) focused on the competencies of processing, evaluating, and producing general media messages. Media literacy was later conceptualized specifically for different forms of media, particularly the Internet and digital media. Examples of media literacy types include new media (Jenkins, 2009; Koc & Barut, 2016), digital (Eshet, 2004), social media (Cho et al., 2022; Tandoc et al., 2021), news (Ashley et al., 2013), information (Chen et al., 2021), and media and information (Yang et al., 2021)

literacy. In the age of digitalized media convergence fueled by ever-changing technologies, a strong consensus has emerged for the need for “a conceptual convergence” (Wuyckens et al., 2022, p. 170) of media literacy, and scholars and the United Nations Educational, Scientific and Cultural Organization have proposed incorporating the various literacies within an integrative framework. Consistent with this proposal, “media literacy” is used as an umbrella term in this study for the literacy concepts listed above. More specifically, media literacy in the context of countering misinformation refers to the ability or skills that enable audiences to diagnose the truthfulness of information they encounter in today’s media environments. A person with adequate media literacy in this regard should be able to accurately assess the credibility of the information, discern misinformation from true information, and engage in further actions that reduce the spread of misinformation (i.e., not sharing misinformation).

The Role of Media Literacy in Combating Misinformation

In light of unprecedentedly widespread misinformation and its substantial adverse outcomes in recent years, the role of media literacy in combating misinformation has attracted growing academic attention. The theoretical underpinning of media literacy lies in inoculation theory, which is a classic persuasion theory concerned with building resistance to persuasive messages through preemptive actions in a manner similar to the action of vaccination against certain viruses (McGuire, 1964). An inoculation message typically comprises two key elements: a *forewarning* and *counterarguing* that activate two persuasion mechanisms threat and refutational preemption, respectively. The *forewarning* element focuses on disclosing the potential threat of a persuasive attack to message receivers, which results in receivers’ heightened motivation to take necessary action required to strengthen their existing beliefs. The *counterarguing* element provides message receivers with specific information that can be employed to refute the respective persuasive attack or techniques that can be used to increase their resistance to subsequent persuasive attempts in general

(Pfau et al., 2005; Traberg et al., 2022). In the past decades, the inoculation method has been employed to improve message receivers' resistance to persuasive attempts concerning a broad range of issues such as anti-smoking (Pfau et al., 1992), genetically modified organisms (Wood, 2007), and native advertising (Amazeen, 2021). A meta-analysis of 41 studies on the effectiveness of inoculation revealed that inoculation treatments were effective in inducing resistance to persuasion attempts (Banas & Rains, 2010).

This approach has also been applied to develop inoculation treatments to reduce susceptibility to misinformation, for example, alerting people of the potential threat of misinformation and exposing people to a small dose of misinformation with correction, corresponding to the *forewarning* and *counterarguing* elements, respectively. Banas and Miller (2013) examined the inoculation effects of fact- versus logic-based messages, which used either example argumentation or critical reasoning strategy to justify the claims. They found that both messages induced resistance toward conspiracy claims, with the fact-based message being more effective than the logic-based one. Roozenbeek et al. (2022a) further discussed two types of inoculation against misinformation: issue-based inoculation and technique-based inoculation. Issue-based inoculation provides corrective information about specific issues, whereas technique-based inoculation focuses on alerting people to the persuasion techniques used in misinformation and teaching them to discern true from false information. These inoculation methods—particularly the technique-based inoculation approach—are aligned with the definition of media literacy. For example, Roozenbeek et al. (2022a) designed the game *Bad News* to inform the public of common manipulation tactics such as conspiracy and discrediting that are used in fake news. They found that playing this game as a media literacy intervention effectively inoculated people against real-world fake news that employed such tactics.

Media literacy interventions in the context of countering misinformation¹—defined as educational programs that equip people with necessary media literacy skills to improve their resilience to misinformation—may impact a variety of outcomes related to misinformation. Evidence regarding perceptual outcomes from the United States, Sweden, Greece, Germany, and Poland showed that media literacy interventions reduced misinformation credibility (Roozenbeek et al., 2020; Scheibenzuber et al., 2021). For example, college students’ perceptions of misinformation credibility decreased significantly after they completed a training course on fake news discernment skills (Scheibenzuber et al., 2021). However, results are less optimistic for interventions implemented on social media platforms. Scholars observed that exposure to news literacy tweets composed of text and infographics had diverging effects on different issues: the intervention decreased the perceived credibility of misinformation about flu vaccines but had no impact for GMO food and raw milk safety (Tully et al., 2020; Vraga et al., 2021).

Media literacy interventions are known to improve individuals’ ability to discern misinformation from true information (Apuke et al., 2022; Al Zou’bi, 2021).

Correspondingly, media literacy training further promotes information verification behaviors

¹ Interventions for countering misinformation can be classified into two categories based on the intervention timing: prebunking and debunking, which occur before and after exposure to misinformation, respectively. The prebunking strategy “attempts to anticipate misinformation by making potential targets aware of its existence, and providing them the necessary knowledge to recognize and counterargue it” (Bertolotti & Catellani, 2023, p.2), whereas debunking involves retrospective corrections and refutations of misinformation. Media literacy interventions fall into the prebunking category as they preemptively inoculate people against misinformation. More specifically, they are considered as technique-based inoculation, which focuses on alerting people to the persuasion techniques used in misinformation and teaching them to discern true from false information. In contrast, there are other prebunking strategies such as issue-based inoculation that provides corrective information about specific issues. For example, in Amazeen et al.’s (2022) experimental study, they exposed participants to a Facebook post from Dr. Fauci in which he refuted the Covid-19 vaccine misinformation with evidence supporting the safety of the vaccine as a prebunking strategy.

and ensures that people can more accurately distinguish true from false information (Domgaard & Park, 2021; Moore & Hancock, 2022). For example, Zhang et al. (2022) exposed participants to an 8-week social media literacy program and concluded that the participants' ability to spot and verify fake news significantly improved after the intervention. In addition to mitigating the negative consequences of exposure to misinformation at the individual level, media literacy interventions have been shown to be effective in containing the spread of misinformation at the societal level by reducing misinformation sharing behaviors (Apuke et al., 2022; Zhang et al., 2022).

In summary, most studies have found a positive association between media literacy interventions and resilience to misinformation, but a few studies reported insignificant effects of media literacy interventions. A meta-analysis that aggregates numerical results across studies would provide a comprehensive view of the role of media literacy interventions in reducing misinformation while accounting for the large variances in samples and study characteristics. In addition to assessing the association between media literacy interventions and the overall outcomes, we investigated the effect of media literacy interventions on specific outcomes including misperceptions, discernment, and sharing. Belief in misinformation taps into the cognitive layer of outcomes, while misinformation discernment and sharing belong to the behavioral layer of outcomes. By dissecting the outcomes, we expected to reveal nuances in the effects of media literacy interventions that could shed light on current media literacy education practices. Based on the above discussions, the following hypothesis was proposed.

Hypothesis 1: *Media literacy interventions have a positive impact on resilience to misinformation (i.e., reducing misperceptions, improving misinformation discernment, and reducing misinformation sharing).*

Potential Moderating Factors

The literature on intervention design in general and media literacy interventions in particular has identified several key factors that may affect the effectiveness of intervention programs. For example, in Sidani's (2017) overview of health interventions, she noted that an intervention's goal, mode of delivery, and dose are critical to whether the intervention can produce the desired outcomes. She also suggested that the intervention outcomes should be assessed as to whether the effects take place immediately and can last over time.

Additionally, much research has documented the importance of adapting interventions to the cultural context where the interventions will be implemented and to the individual characteristics of the target population (Castro et al., 2010; Kreuter et al., 2003; Manson, 2020). Moreover, the recent developments in inoculation research have identified that the inoculation processes and effects are subject to the influence of many factors such as features of the inoculation treatment, target that receives the inoculation message, as well as contextual factors (Compton et al., 2021). Thus, we proposed an integrated model of intervention design in the context of media literacy programs by incorporating all these factors in light of inoculation theory (see Figure 1) and discussed what role these factors might play in the inoculation processes related to countering misinformation.

[Figure 1 Here]

Intervention Design and Evaluation Factors

How a media literacy intervention is implemented may play a role in shaping its effectiveness. Prior meta-analyses have examined various factors of intervention design and evaluation, such as the intervention medium (Vahedi et al., 2018), the dose of intervention sessions (Jeong et al., 2012), and the timing for measuring the effect (Xie et al., 2019), as potential moderators of intervention effectiveness in contexts other than misinformation. We consider the three moderators as theoretically relevant and practically important in the

context of combating misinformation and therefore include these moderators in the meta-analysis.

As part of literacy education programs that aim to improve citizens' ability to recognize misinformation, relevant education materials have been developed in various formats, including infographics (e.g., Domgaard & Park, 2021), videos (e.g., Roozenbeek et al., 2022b), and games (e.g., Yang et al., 2021). Different media formats engage with different physical senses (e.g., vision and hearing) and they vary in the way that they communicate, which may lead to differences in message processing and subsequent inoculation effects (Pfau et al., 2000). Previous research found that inoculation as a preemptive strategy was equally effective with material in video and print format in the case of crisis communication (Dillingham & Ivanov, 2017). In the context of media literacy education, a study showed that using visual multimedia materials to facilitate a face-to-face literacy education course significantly improved participants' knowledge about fake news and the pre-post increase was greater than that without such materials (Zhang et al., 2022). These findings are consistent with the theoretical explanation that media content in multiple modalities (e.g., text and visual) generates better learning outcomes than that in a single thin modality such as text (Clark, 2001). Moreover, interactivity has been shown to be an important factor in enhancing learning effectiveness (Evans & Gibbons, 2007). Interactive media (e.g., game) may differ in their capacity in inoculation from non-interactive media (e.g., video and print), given the different levels of cognitive engagement associated with using interactive versus non-interactive media (Young et al., 2012). Compton (2024) notes that an interactive game that involves inoculation against fake news is an example for the active (versus passive) inoculation approach, that is, users will actively generate counter-arguments to the attacking messages during the processes of playing the game, and this enhanced cognitive engagement will result in higher inoculation efficacy. Relevant empirical

studies found that playing literacy education games elicited the inoculation effect and reduced participants' credibility perceptions of fake news (Roozenbeek et al., 2020; Yang et al., 2021). Hence, we expected that media literacy programs in various formats and at different levels of interactivity would result in divergent outcomes in countering misinformation. Therefore, the following hypothesis was proposed.

Hypothesis 2: *The effect of media literacy interventions on resilience to misinformation varies by the modality of intervention delivery.*

The dose of media literacy intervention sessions may impact the effectiveness of interventions. In their meta-analysis of 51 media literacy interventions, Jeong et al. (2012) observed that the number of sessions was positively correlated with intervention effectiveness. To address misinformation, some media literacy interventions involved a single session (Qian et al., 2022), whereas others implemented multiple sessions (e.g., Scheibenzuber et al., 2021). According to inoculation theory, after an initial inoculation treatment message is sent to a message receiver, a booster message that contains a warning reminder about the incoming attacks of opposing arguments and repeated refutations would further bolster the attitude induced by the initial message (Compton & Pfau, 2005; Ivanov et al., 2018). However, no empirical studies have investigated whether the effect of media literacy intervention is influenced by the number of sessions in an intervention program against misinformation. To fill this research gap and tackle this practically important issue, we proposed the following hypothesis in line with the finding of Jeong et al. (2012).

Hypothesis 3: *The effect of media literacy interventions on resilience to misinformation varies by the dose of intervention sessions.*

Whether media literacy programs have only an immediate effect or a lasting effect is a critical issue for educators. Prior research on deviant behaviors suggested that the effectiveness of media literacy programs about smoking and weight concerns diminished

over time (Pfau et al., 1992; Wade et al., 2003). Nevertheless, a meta-analysis of 23 studies showed no significant difference between the immediate and delayed effects of media literacy interventions on deviant behaviors (Xie et al., 2019). The literature suggests that an inoculation treatment can be more effective at a later time as individuals may need time to take in the inoculation and generate arguments to defend their attitudes (Banas & Rains, 2010; McGuire, 1964). However, recent research notes that people's memory may decay quickly after they receive an inoculation treatment, so the inoculation effect may not sustain if no other actions are taken to reinforce the learnings (Capewell et al., 2024). In the context of countering misinformation, a few studies have measured the intervention effect in a follow-up test of interventions (e.g., Roozenbeek et al., 2022b), whereas most studies have focused on the effect immediately after the interventions (e.g., Domgaard & Park, 2021). Therefore, we applied a meta-analytic method to investigate whether the effect of media literacy interventions would differ by the timing of measurement, as described by the following hypothesis.

Hypothesis 4: *The effect of media literacy interventions on resilience to misinformation varies by the timing for measuring the effect.*

Culture: Uncertainty Avoidance

The widespread dissemination of misinformation has become a global issue, and the growing body of research on media literacy and misinformation reduction has encompassed many geographic locations and cultural contexts, including Germany (Scheibenzuber et al., 2021), Ghana (Adjin-Tettey, 2022), Nigeria (Apuke et al., 2022), South Korea (Hwang et al., 2021), and the United States (Moore & Hancock, 2022; Vraga et al., 2021; 2022). Some cross-cultural studies have found divergent results in different places. For example, Guess et al. (2020) implemented a digital media literacy intervention involving tips on misinformation discernment in two countries. They found that the intervention was more effective among

American respondents than among their Indian counterparts both immediately after the intervention and after a delay. Therefore, the variation in results from prior studies may be partially attributed to cultural differences.

Hofstede's (1991) cultural dimension framework provides an appropriate angle for studying the role of national cultural traits in various social phenomena. Among cultural dimensions, uncertainty avoidance—defined as “the extent to which the members of a culture feel threatened by uncertain or unknown situations” (Hofstede, 1991, p. 113)—is closely related to uncertainty management, information processing, and security needs. This cultural dimension has thus been examined in misinformation studies to explain cross-cultural differences. For example, people in cultures that feature higher levels of uncertainty avoidance reported higher perceived exposure to misinformation (Arrese, 2022). Moreover, exposure to misinformation had a stronger association with information avoidance in high uncertainty avoidance cultures (e.g., South Korea) than in low uncertainty avoidance cultures such as the United States (Kim et al., 2020). Besides, the effect of intervention for online information may be moderated by cultural differences in uncertainty avoidance as this cultural dimension is related to (in)tolerance for ambiguity and the need for structure in the process (Vishwanath, 2003). In the context of media literacy interventions, a strong tendency toward uncertainty avoidance may be linked to a strong need to absorb new knowledge and skills that are useful for reducing uncertainty in information processing. Moreover, people in high uncertainty avoidance cultures may be more motivated to use their information and media skills to authenticate information encountered in media environments than those in low uncertainty avoidance cultures.

Therefore, we expected the association between media literacy and resilience to misinformation to be stronger for high uncertainty avoidance cultures than for low uncertainty avoidance cultures.

Hypothesis 5: *The effect of media literacy interventions on resilience to misinformation varies by the degree of uncertainty avoidance of the cultural context.*

Sample Type

Studies on media literacy and misinformation have utilized various types of samples, including college students (e.g., Scheibenzuber et al., 2021), representative adult populations (e.g., Vraga et al., 2021), samples from crowdsourcing platforms like MTurk and Prolific (Tully et al., 2020), and special population segments such as older adults (Moore & Hancock, 2022). The differential demographic characteristics of the samples may play a role in the effect sizes in meta-analyses (Walter et al., 2018); this may also be the case in the relationship between media literacy and misinformation-related outcomes. For example, a study of adults in China showed that age was positively related to misperceptions about COVID-19, but education was a negative predictor (Su et al., 2022). Moore and Hancock (2022) observed that older adults were more vulnerable to the influence of fake news than young adults because of insufficient digital literacy among older adults. Moreover, the psychology literature has noted that individuals' need for cognition, a critical dimension of media literacy (Ashley et al., 2013), varies across life stages (Cacioppo et al., 1996). This indicates that the tendency to engage in the effortful processing of information varies by age and may affect misinformation-related perceptions and behaviors such as authentication. Cameron (2009) posits that "[I]noculation theory may be most useful for those practitioners whose patient population includes children and young adults" (p. 314), which warrants empirical validation. Accordingly, we expected the inoculation effect of media literacy interventions on resilience to misinformation to vary by sample type as a result of variations in demographic characteristics. For example, college students are younger and more educated than non-students. Therefore, we proposed the following hypothesis.

Hypothesis 6: *The effect of media literacy interventions on resilience to misinformation varies by sample type.*

Methods

Literature Search and Screening

We conducted systematic searches in five electronic databases (i.e., PsycINFO, EBSCO, Web of Science, ProQuest (including Dissertations and Theses), and Google Scholar) to retrieve studies that examined the effect of media literacy interventions on misinformation-related outcomes. The searches were conducted from April 7 to April 11, 2022 with the following search strings: “misinformation” OR “disinformation” OR “rumor” OR “fake news” OR “false” AND “media literacy” OR “information literacy” OR “news literacy” OR “literacy program” OR “literacy education.” These search strings were chosen in light of previous systematic review and meta-analysis on relevant topics (Jeong et al., 2012; Yu et al., 2023). The searches generated 575 articles after the removal of duplicates.

We then assessed the articles for eligibility against our inclusion criteria. Experimental studies that examined the effect of a media literacy intervention on misinformation-related outcomes were included in the meta-analysis. Studies that did not include persuasion outcome variables related to misinformation—such as misinformation belief, misinformation sharing, and misinformation discernment—were excluded. Studies that adopted a prebunking or inoculation strategy to combat misinformation but used issue-specific rebuttal rather than a technique-focused approach were excluded. In addition, included studies should have reported appropriate statistics for calculating the effect size (i.e., means, standard deviations, and sample sizes). For articles that did not provide sufficient statistical information for effect size computation, we contacted the corresponding authors to request the information. Articles for which we were unable to calculate the effect sizes due to lack of information were excluded. We also checked the publication record of prolific

scholars in this area and identified several eligible studies for inclusion. Another round of search was conducted in February 2024, and 4 additional articles were identified. Ultimately, 29 articles that reported the results of 49 experimental studies ($N = 81,155$) were included in the analysis (see Figure S1 in the Supplementary Material for the PRISMA flow diagram).

Effect Size Extraction

Effect size estimates of the effects of media literacy interventions on misinformation-related outcomes were extracted from the included studies. For post-measure-only between-subjects intervention studies, the standardized mean difference (d) in the outcome variables between the treatment (i.e., media literacy intervention) and the control (i.e., no intervention) groups was calculated based on the means and standard deviations of the outcome variables and the sample sizes of the groups (Borenstein et al., 2009). For one-group within-subjects design intervention studies (e.g., Scheibenzuber et al., 2021), the effect sizes were calculated based on the means of the pre- and post-intervention measures and the pooled pre- and post-intervention standard deviations (Cumming, 2012). For studies with a Solomon design (i.e., pre- and post-measures for the treatment and control groups), the effect sizes were computed based on the means of the post-intervention outcomes, the pooled pre-intervention standard deviations of the groups, and the sample sizes of the groups (Morris, 2008).

For studies that involved multiple misinformation-related outcomes, the effect size for each outcome was recorded as a separate entry in the meta-analytic dataset. A positive d indicated a positive effect of media literacy intervention on resilience to misinformation. For outcome variables that were in the opposite direction—such as belief in misinformation (e.g., Lee et al., 2022; Su et al., 2022)—the direction of the effect size estimate was reversed to ensure these variables were consistent with other entries in the dataset. Specifically, a positive d for belief in misinformation signified that the media literacy intervention reduced belief in misinformation and improved resilience to misinformation.

Moderator Coding

Moderators were coded by two independent coders based on the information provided in the articles (see Table S1 in the Supplementary Material for the information of the moderators for all the articles). The intercoder reliability for all the moderators was satisfactory (Krippendorff's $\alpha > .80$). The modality of the intervention materials was coded as follows: text and picture/infographic, video, and game. We coded the dose of intervention sessions as either "one" or "multiple," with "multiple" signifying that the intervention included more than one session. For uncertainty avoidance, the coders first recorded the country/region of the study and then obtained the uncertainty avoidance score for each location from Hofstede's website.² Studies conducted in locations with scores of 50 or below and above 50 were coded as low and high, respectively, in uncertainty avoidance. Sample composition was coded as follows: student sample, crowdsourcing sample (i.e., samples recruited from crowdsourcing platforms such as MTurk and Prolific), convenience sample, and other (e.g., older adults).

Analytic Procedure

We performed all the analyses with the *metafor* package in R. We adopted a multilevel meta-analytic approach to address the statistical dependency of multiple effect sizes generated from a single sample by synthesizing the effect sizes while accounting for variances at two levels (i.e., level 1: participant level; level 2: sample level; Cheung, 2019; Van den Noortgate et al., 2015). Next, we conducted a series of subgroup analyses to test the moderating effects, with a significant F-test result suggesting significant differences among the categories. We also calculated average effect sizes for each level of moderator.

Results

Effect Size Estimation

² <https://www.hofstede-insights.com/>

The results showed that media literacy interventions significantly increased resilience to misinformation ($d = 0.60$, 95% CI [0.33, 0.86], $k = 117$, $se = 0.13$, $t = 4.51$, $p < .0001$). Specifically, media literacy significantly reduced belief in misinformation ($d = 0.27$, 95% CI [0.18, 0.37], $k = 53$, $se = 0.05$, $t = 5.79$, $p < .0001$) and sharing of misinformation ($d = 1.04$, 95% CI [0.10, 1.98], $k = 9$, $se = 0.41$, $t = 2.55$, $p = .03$), and promoted misinformation discernment ($d = 0.76$, 95% CI [0.38, 1.15], $k = 49$, $se = 0.19$, $t = 3.99$, $p < .001$). According to Cohen (1992), the cut-off values for small, medium, and large effect sizes are $d = 0.20$, 0.50, and 0.80, respectively. Therefore, media literacy interventions had a medium-sized effect on resilience to misinformation. Specifically, media literacy interventions had a large effect on reducing misinformation sharing, a medium-to-large effect on enhancing misinformation discernment, and a small effect on reducing misinformation belief.

Moderator Analysis

The Q test revealed significant heterogeneity in the sample, $Q(116) = 6750.07$, $p < .0001$. The value of I^2 for each of the three levels indicates the amount of heterogeneity variance attributable to each level. The sampling error variance on level 1 (i.e., participant level) accounted for only 0.17% of the total variance in the data ($I^2_{level\ 1} = 0.17$). The percentage of the within-study variance was 35.80% ($I^2_{level\ 2} = 35.80$) and that of the between-study variance was 64.03% ($I^2_{level\ 3} = 64.03$). This suggests that there is substantial heterogeneity between the studies included in the meta-analysis and potential moderators may have accounted for the variance. Therefore, we performed a series of analyses on the proposed moderators to further investigate this issue.

We first investigated whether the effect of media literacy interventions on misinformation-related outcomes varied by intervention design factors including the intervention format, the dose of intervention sessions, and the timing for measuring the effect. The results showed that the modality of the intervention materials did not affect the

effectiveness, $F(2, 88) = 0.56, p = .57$. Neither did the timing (immediate vs. delayed) for measuring the effect make a difference to the outcomes, $F(1, 109) = 0.02, p = .89$. Studies with multiple intervention sessions ($d = 1.93, 95\% \text{ CI } [0.93, 2.92], k = 26, se = 0.48, t = 3.97, p < .001$) reported a stronger average effect than those involving only one session ($d = 0.26, 95\% \text{ CI } [0.18, 0.34], k = 91, se = 0.04, t = 6.70, p < .0001$), $F(1, 115) = 37.27, p < .0001$. It is worth noting that among the three modalities employed by the interventions, only studies using game for literacy education found a significant average effect of media literacy interventions ($d = 0.31, 95\% \text{ CI } [0.19, 0.42], k = 40, se = 0.06, t = 5.40, p < .0001$).

We observed that the relationship between media literacy interventions and resilience to misinformation varied by cultural context, $F(1, 87) = 13.19, p < .001$. Studies conducted in locations featuring high levels of uncertainty avoidance ($d = 1.46, 95\% \text{ CI } [0.63, 2.29], k = 35, se = 0.41, t = 3.58, p = .001$) reported a larger average effect size than those in contexts featuring lower levels of uncertainty avoidance ($d = 0.22, 95\% \text{ CI } [0.14, 0.30], k = 54, se = 0.04, t = 5.56, p < .0001$).

We also found that the effect of media literacy interventions on resilience to misinformation differed across the types of samples, $F(3, 103) = 15.43, p < .0001$. The effect was stronger in studies using student samples ($d = 2.17, 95\% \text{ CI } [1.06, 3.27], k = 24, se = 0.53, t = 4.06, p < .001$) than in those using samples recruited from online crowdsourcing platforms ($d = 0.29, 95\% \text{ CI } [0.21, 0.37], k = 59, se = 0.04, t = 7.35, p < .0001$) and convenience samples ($d = 0.32, 95\% \text{ CI } [0.22, 0.42], k = 10, se = 0.04, t = 7.18, p < .0001$).

[Table 1 Here]

We also performed moderator analyses on misinformation belief ($Q(52) = 1005.31, p < .0001, I^2_{level 1} = 1.40, I^2_{level 2} = 32.92, I^2_{level 3} = 65.68$) and discernment ($Q(48) = 2849.44, p < .0001, I^2_{level 1} = 0.39, I^2_{level 2} = 3.96, I^2_{level 3} = 95.65$) following the same procedure above.

The outcome variable misinformation sharing was not examined separately as the sample size

involving misinformation sharing was small ($k = 9$). We found that intervention dose significantly moderated the efficacy of media literacy interventions in reducing misinformation beliefs, $F(1, 51) = 4.79, p = .03$. Specifically, media literacy programs with multiple sessions ($d = 0.58, 95\% \text{ CI } [0.17, 0.998], k = 9, se = 0.18, t = 3.23, p = .01$) were more effective than those with one session ($d = 0.23, 95\% \text{ CI } [0.13, 0.33], k = 44, se = 0.05, t = 4.53, p < .0001$) in curbing misinformation beliefs. Moreover, media literacy interventions in higher uncertainty avoidance cultures ($d = 0.44, 95\% \text{ CI } [0.24, 0.64], k = 13, se = 0.09, t = 4.80, p < .001$) had stronger effects than those in lower uncertainty avoidance cultures ($d = 0.10, 95\% \text{ CI } [0.04, 0.16], k = 18, se = 0.03, t = 3.29, p = .004$). The effectiveness of media literacy interventions also varied by sample type, $F(3, 40) = 3.71, p = .02$. Media literacy interventions had stronger effects in misinformation belief reduction among college students ($d = 0.88, 95\% \text{ CI } [-0.04, 1.79], k = 6, se = 0.36, t = 2.47, p = .06$) than with samples recruited from crowdsourcing platforms ($d = 0.28, 95\% \text{ CI } [0.14, 0.43], k = 24, se = 0.07, t = 4.11, p < .001$) and other sources ($d = 0.08, 95\% \text{ CI } [-0.05, 0.21], k = 6, se = 0.05, t = 1.58, p = .17$).

Similar patterns were observed regarding the misinformation discernment outcome such that intervention dose, the level of uncertainty avoidance, and sample type were significant moderators of the effect of media literacy interventions. Intervention programs with multiple sessions ($d = 2.32, 95\% \text{ CI } [1.58, 3.05], k = 9, se = 0.32, t = 7.29, p < .0001$) were more effective than those with a single session ($d = 0.32, 95\% \text{ CI } [0.24, 0.39], k = 40, se = 0.04, t = 8.48, p < .0001$), $F(1, 47) = 107.36, p < .0001$. Participants in higher uncertainty avoidance places ($d = 1.74, 95\% \text{ CI } [0.77, 2.71], k = 11, se = 0.43, t = 4.00, p = .003$) scored higher in misinformation discernment after receiving an media literacy intervention than their counterparts in lower uncertainty avoidance places ($d = 0.32, 95\% \text{ CI } [0.21, 0.43], k = 34, se = 0.05, t = 5.93, p < .0001$), $F(1, 43) = 18.65, p < .0001$. Additionally, media literacy

interventions were more effective in increasing misinformation discernment ability among students ($d = 2.32$, 95% CI [1.58, 3.05], $k = 9$, $se = 0.32$, $t = 7.29$, $p < .0001$) than among participants recruited on online crowdsourcing platforms ($d = 0.32$, 95% CI [0.25, 0.38], $k = 32$, $se = 0.03$, $t = 9.61$, $p < .0001$).

[Table 2 Here]

[Table 3 Here]

Publication Bias Diagnosis

Publication bias is a common caveat that concerns the results of meta-analyses. It refers to the tendency that empirical studies with significant test results and larger effect sizes are more likely to pass the peer review process and eventually to be published, resulting in an overestimation of the effect size estimates by meta-analyses compared to the actual effect in the real-world (Rosenthal, 1979). Consistent with the recommended practices of previous meta-analyses, we applied the trim-and-fill procedure to examine potential publication bias (Duval & Tweedie, 2000). We first created a funnel plot to visualize the distribution of the effect sizes and then assessed the number of studies with small samples and effect sizes that may have been missing to achieve distribution symmetry. The results showed that the estimated number of missing studies on the left side was zero ($se = 1.95$), indicating no evidence for publication bias that may affect this study.

Discussion

Identifying ways to mitigate increasingly prevalent misinformation is a critical issue facing governments, news agencies, educators, and the general public. A recommended solution is to improve people's ability to discern true from false information through appropriate media literacy education programs. This meta-analysis of 49 experimental studies that examined the effectiveness of media literacy interventions in combating misinformation suggests that media literacy interventions generally improve resilience to misinformation

with a medium-sized effect ($d = 0.60$). Media literacy interventions significantly reduce misinformation beliefs, promote misinformation discernment, and decrease misinformation sharing. Furthermore, the effects are moderated by the number of intervention sessions, the cultural context, and the sample source. The theoretical and practical implications of the findings are discussed as follows.

To the best of our knowledge, this study is the first meta-analysis of the utility of media literacy interventions in countering misinformation and provides strong evidence for the effectiveness of these intervention programs. The average effect size ($d = 0.60$) revealed by this study is comparable to that reported by a previous meta-analysis on the effect of media literacy interventions across domains ($d = 0.37$; Jeong et al., 2012) and a meta-analysis on the effect of inoculation ($d = 0.43$, Banas & Rains, 2010). This lends support to the effectiveness of the technique-based inoculation approach to combating misinformation and thus contributes to the inoculation literature in this context. This study also enriches our understanding of inoculation theory by showing that the effectiveness of technique-based inoculation may depend on the specific outcome, cultural context, delivery method, and target population, suggesting that a one-size-fits-all approach for inoculation is not optimal in the fight against misinformation.

This study reveals that media literacy interventions are effective in inducing changes in various outcomes related to misinformation. Specifically, the interventions have a small effect on reducing misperceptions ($d = 0.27$), a medium-to-large effect on improving misinformation discernment ($d = 0.76$), and a large effect on decreasing misinformation sharing ($d = 1.04$). This suggests that the impact of media literacy programs is far-reaching and ranges from reducing the harmful consequences of exposure to misinformation at the individual level to containing the spread of misinformation at the community level. Interestingly, the magnitude of the effect on various outcomes may not be equal; media

literacy interventions appear to have a larger effect on improving misinformation discernment and mitigating misinformation sharing than on directly reducing misperceptions. Hence, given the multi-faceted, far-reaching effectiveness of media literacy interventions in combating misinformation, we recommend that media literacy programs that offer trainings on false information discernment skills should be provided to the general public to enhance their resilience to misinformation.

Although converging evidence has been identified for the utility of media literacy programs in countering misinformation, how the programs are designed and implemented may make a difference to its effectiveness. This raises an important question for media literacy scholars: how should a highly effective media literacy program be designed? Consistent with the finding of Jeong et al. (2012) that the number of sessions is positively associated with intervention effectiveness, our meta-analysis reveals that media literacy intervention programs that involve multiple intervention sessions ($d = 1.93$) have stronger effects than those with only one session ($d = 0.26$). Therefore, based on the meta-analytic evidence, we believe that the advantage of using multiple sessions over a one-off approach exists not only in the domain of countering misinformation but may also be extended to other domains as a general principle for intervention design. Future studies are recommended to investigate the effect of one versus multiple intervention sessions in real-world settings to further validate this finding. However, it should be noted that multiple-session interventions can lead to a higher level of resistance or attrition, which diminishes the efficacy of widespread inoculation in reality. It would be necessary to identify how multiple sessions can be organized to retain participants and reduce resistance. For example, the optimal number of sessions in one intervention program, the interval length between sessions, and whether sessions should be repeated or sequential can be explored. Elucidating the roles of these factors can provide useful insights into the design of future media literacy programs.

As mentioned previously, the prevalence of misinformation is a global concern, and media literacy programs have been implemented in many places worldwide. However, the effectiveness of these programs is not uniform across cultural contexts. Guided by Hofstede's (1991) cultural dimensions framework, this study finds that the effect of media literacy interventions is stronger in high versus low uncertainty avoidance contexts, consistent with previous studies (Kim et al., 2020). People in high uncertainty avoidance contexts reported higher perceived misinformation exposure than those in low uncertainty avoidance contexts (Arrese, 2022), signifying that those in high uncertainty avoidance contexts may be more alert to the prevalence of misinformation and more willing to acquire essential skills to help them authenticate the information to reduce uncertainty. This may lead to the enhanced effectiveness of media literacy programs in high uncertainty avoidance cultures ($d = 1.46$). Given the considerable psychological need to reduce information uncertainty in high uncertainty avoidance cultures such as South Korea and Japan, we suggest that media literacy programs be widely provided to the general public. Such programs are expected to be well-received and highly effective in these countries. In contrast, although media literacy programs can also increase resilience to misinformation in low uncertainty avoidance cultures such as the United States and Hong Kong, their effect is smaller ($d = 0.22$) in these places. We urge more academic effort toward identifying intervention design factors that can boost the effectiveness of media literacy programs in low uncertainty avoidance contexts in particular.

Moreover, this study shows that the effectiveness of media literacy interventions in improving resilience to misinformation is conditioned by sample characteristics. Specifically, the interventions are more effective among college students ($d = 2.17$) than among adult samples from crowdsourcing platforms ($d = 0.29$)—a finding consistent with our prediction that young, highly educated participants are more receptive to media literacy education

programs. This raises a strong warning about the vulnerable position of older adults with low social economic status (e.g., low income, low education level) in the fight against misinformation. It is sadly possible that even if the disadvantaged have equal access to these literacy education programs, the effectiveness of these programs may be lower in disadvantaged populations versus more advantaged groups because of the insufficient knowledge base and literacy skills of the disadvantaged (Moore & Hancock, 2022; Seo et al., 2021). Therefore, special care should be taken in designing media literacy programs tailored to older adults with low education levels to ensure they do not lag behind other population segments in today's information society. It should be noted that most of the studies in our sample targeted college students or adult populations. More research is thus called for to explore the specific factors in intervention design that can enhance the receptivity of vulnerable populations (e.g., elderly people, less educated people) to media literacy interventions that will help reduce the harm of rampant misinformation.

In contrast to previous empirical studies that have concluded that multimodal and interactive media enhance learning outcomes (Clark, 2001; Roozenbeek et al., 2020; Yang et al., 2021; Zhang et al., 2022), this meta-analysis does not find a significant moderating effect of the modality of intervention materials. The non-significant result may be explained by the characteristics of the intervention materials involved in our dataset. The studies included in this meta-analysis mainly used three media modalities: text and picture/infographic, video, and game. All of these modalities are “richer” than pure text; therefore, no significant difference was detected among these three “rich” media. Future research is recommended to identify the most effective modalities for delivering fake news discernment education programs targeted at specific population segments such as children and older adults. Additionally, our meta-analysis does not offer evidence that the timing for measuring the effect is a significant moderator of intervention effectiveness. This finding echoes that of a

previous meta-analysis on unhealthy behaviors (Xie et al., 2019). However, given the relatively small number of studies ($k = 18$) in our sample that investigated delayed effects, we are reluctant to derive a definitive conclusion on this matter. Understanding whether the learning effect of literacy education programs diminishes over time is of vital importance to educators. More research is recommended to demystify the duration of the effect of media literacy interventions on misinformation-related outcomes; this line of research is expected to make a valuable contribution to the literature on media effects and guide relevant media literacy practices.

Notably, our study has some limitations that should be considered when interpreting the findings. The first is a general limitation of the meta-analytic method. Meta-analysis is a quantitative synthesis of existing research on a certain topic, so the results of meta-analyses may be affected by the availability of the statistical information of eligible studies and the quality of the primary studies included in the sample (Matt & Cook, 2019). The second limitation stems from the relatively small amount of literature in the field. Two research groups are particularly prolific on this topic and have contributed 16 of the 29 articles in the sample of this meta-analysis. We appreciate the substantial contributions of these scholars to this important line of research, but our meta-analytic findings may be subject to the influence of the large portion of studies in our sample conducted by the two research groups. We expect that more resources will be provided to media literacy education programs and relevant academic studies so that a more diverse group of scholars will be able to engage in this line of research. Third, we used uncertainty avoidance, one of the five cultural dimensions identified by Hofstede (1991), as a macro-level indicator of a national trait. However, individual levels of this dimension may vary within a country. Therefore, the scores of uncertainty avoidance are rough approximations of the levels of this dimension of the countries involved. Individual-level research on the role of uncertainty avoidance in

countering misinformation in which uncertainty avoidance can be viewed as a psychological trait is needed. Fourth, belief in misinformation is a cognitive outcome and misinformation discernment and sharing are behavioral outcomes. It is possible that these outcomes relate to each other; however, we were unable to test that with this meta-analysis. It is recommended that future research explore the interplay among the various misinformation-related outcomes and the potential mediating mechanisms. Last but not the least, despite the meta-analytic evidence for its overall positive effect in improving resilience to misinformation, extra caution needs to be taken when we apply the findings into real-world settings. For one thing, some practical elements may affect the implementation and outcome of media literacy programs in real-world settings such as participant recruitment and resource availability. For another, media literacy interventions can have negative spill-over effects. For example, misinformation warnings may prime people in terms of the heightened prevalence of fake news, resulting in an increased distrust in real news (van der Meer et al., 2023; Van Duyn & Collier, 2019).

To conclude, this study provides meta-analytic evidence for the effectiveness of media literacy interventions in combating misinformation. Interventions can reduce beliefs in misinformation, improve misinformation discernment, and decrease misinformation sharing. This study also finds that intervention design, cultural context, and sample characteristics may affect the effectiveness of the interventions. Specifically, interventions with multiple sessions have stronger effects than those with a single session and media literacy interventions are more effective in high versus low uncertainty avoidance cultures and among college students versus adults from crowdsourcing platforms. This study has rich practical implications for media literacy education practices. First, literacy education organizations can cite this study as a piece of solid evidence for the effectiveness of media literacy interventions in countering misinformation to support their solicitation for financial support

for implementing this type of programs. Second, this study finds that a one-size-fits-all approach does not apply to the intervention design in this domain. We recommend that the design of appropriate interventions should be tailored to the needs of the target population as well as the cultural programming of a particular context. Third, when resource allows, multiple-session interventions are recommended as they are more effective than those with only one session, though extra caution should be taken as to how to recruit and retain participants.

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Figure 1 *An Integrated Model of Media Literacy Intervention Design to Mitigate Misinformation*

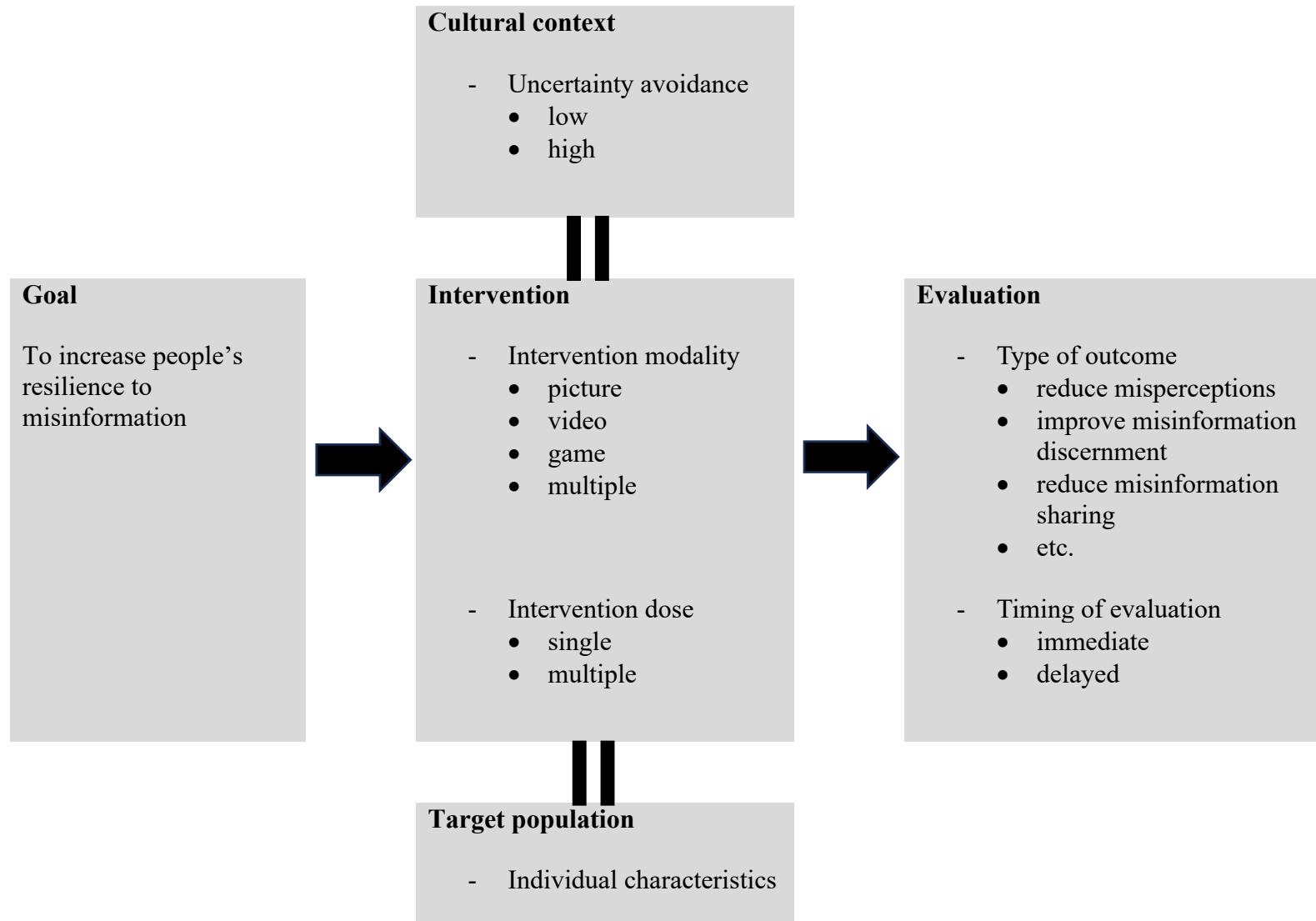


Table 1*Results of Moderator Analyses on Misinformation Mitigation Outcomes ($k = 117$)*

Moderator	Level	k	d	se	t	p	95% CI	Omnibus test
Intervention modality	Picture ¹	18	0.47	0.33	1.41	.18	[-0.23, 1.17]	$F(2, 88) = 0.56, p = .57$
	Video	33	0.73	0.47	1.55	.13	[-0.23, 1.68]	
	Game	40	0.31***	0.06	5.40	<.0001	[0.19, 0.42]	
Intervention dose	One	91	0.26***	0.04	6.70	<.0001	[0.18, 0.34]	$F(1, 115) = 37.27, p < .0001$
	Multiple	26	1.93***	0.48	3.97	<.001	[0.93, 2.92]	
Timing of measurement	Immediate	91	0.62***	0.16	3.92	<.001	[0.31, 0.94]	$F(1, 109) = 0.02, p = .89$
	Delayed	18	0.69*	0.26	2.62	.02	[0.14, 1.25]	
Uncertainty avoidance	Low	54	0.22***	0.04	5.56	<.0001	[0.14, 0.30]	$F(1, 87) = 13.19, p < .001$
	High	35	1.46**	0.41	3.58	.001	[0.63, 2.29]	
Sample type	Student	24	2.17***	0.53	4.06	<.001	[1.06, 3.27]	$F(3, 103) = 15.43, p < .0001$
	MTurk ²	59	0.29***	0.04	7.35	<.0001	[0.21, 0.37]	
	Convenience	10	0.32***	0.04	7.18	<.0001	[0.22, 0.42]	
	Other	14	0.25	0.13	1.88	.08	[-0.04, 0.53]	

Note. 1. “Picture” represents the category in which the experiment used stimulus material in text and picture/infographic. 2. “MTurk” represents the category in which the study used samples recruited from crowdsourcing platforms such as Amazon Mechanical Turk.

* $p < .05$, ** $p < .01$, *** $p < .001$

Table 2

Results of Moderator Analyses on Reducing Misinformation Beliefs ($k = 53$)

Moderator	Level	k	d	se	t	p	95% CI	Omnibus test
Intervention modality	Picture ¹	11	0.10	0.05	2.11	.06	[-0.006, 0.20]	$F(2, 41) = 2.43, p = .10$
	Video	2	-0.03	0.06	-0.55	.68	[-0.73, 0.67]	
	Game	31	0.31***	0.06	5.07	<.0001	[0.19, 0.44]	
Intervention dose	One	44	0.23***	0.05	4.53	<.0001	[0.13, 0.33]	$F(1, 51) = 4.79, p = .03$
	Multiple	9	0.58*	0.18	3.23	.01	[0.17, 0.998]	
Timing of measurement	Immediate	37	0.30***	0.07	4.24	<.001	[0.16, 0.44]	$F(1, 48) = 0.004, p = .95$
	Delayed	12	0.36***	0.02	17.59	<.0001	[0.32, 0.41]	
Uncertainty avoidance	Low	18	0.10**	0.03	3.29	.004	[0.04, 0.16]	$F(1, 29) = 28.71, p < .0001$
	High	13	0.44***	0.09	4.80	<.001	[0.24, 0.64]	
Sample type	Student	6	0.88	0.36	2.47	.06	[-0.04, 1.79]	$F(3, 40) = 3.71, p = .02$
	MTurk ²	24	0.28***	0.07	4.11	<.001	[0.14, 0.43]	
	Convenience	8	0.37***	0.03	10.90	<.0001	[0.29, 0.46]	
	Other	6	0.08	0.05	1.58	.17	[-0.05, 0.21]	

Note. 1. “Picture” represents the category in which the experiment used stimulus material in text and picture/infographic. 2. “MTurk” represents the category in which the study used samples recruited from crowdsourcing platforms such as Amazon Mechanical Turk.

* $p < .05$, ** $p < .01$, *** $p < .001$

Table 3

Results of Moderator Analyses on Misinformation Discernment ($k = 49$)

Moderator	Level	k	d	se	t	p	95% CI	Omnibus test
Intervention modality	Picture ¹	5	1.01	0.69	1.47	.22	[-0.90, 2.92]	$F(2, 36) = 0.74, p = .48$
	Video	27	0.52	0.26	1.99	.06	[-0.02, 1.06]	
	Game	7	0.31***	0.04	6.95	<.001	[0.20, 0.41]	
Intervention dose	One	40	0.32***	0.04	8.48	<.0001	[0.24, 0.39]	$F(1, 47) = 107.36, p < .0001$
	Multiple	9	2.32***	0.32	7.29	<.0001	[1.58, 3.05]	
Timing of measurement	Immediate	43	0.73***	0.19	3.78	<.001	[0.34, 1.12]	$F(1, 46) = 0.89, p = .35$
	Delayed	4	1.27	0.80	1.60	.21	[-1.26, 3.81]	
Uncertainty avoidance	Low	34	0.32***	0.05	5.93	<.0001	[0.21, 0.43]	$F(1, 43) = 18.65, p < .0001$
	High	11	1.74**	0.43	4.00	.003	[0.77, 2.71]	
Sample type	Student	9	2.32***	0.32	7.29	<.0001	[1.58, 3.05]	$F(3, 45) = 33.10, p < .0001$
	MTurk ²	32	0.32***	0.03	9.61	<.0001	[0.25, 0.38]	
	Convenience	2	0.22	0.14	1.66	.35	[-1.49, 1.94]	
	Other	6	0.60	0.39	1.53	.19	[-0.40, 1.60]	

Note. 1. “Picture” represents the category in which the experiment used stimulus material in text and picture/infographic. 2. “MTurk” represents the category in which the study used samples recruited from crowdsourcing platforms such as Amazon Mechanical Turk.

** $p < .01$, *** $p < .001$